

Bilateral Alterations in Isokinetic Strength and Knee Biomechanics During Side-Cutting 1 Year After Unilateral ACL Reconstruction

Huijuan Shi,^{*,†} MD, Shuang Ren,[†] PhD, Hongshi Huang,[†] MD, Hui Liu,^{*,‡} PhD, Zixuan Liang,[†] BS, Yuanyuan Yu,[†] MS, Hanjun Li,^{*} PhD, and Yingfang Ao,^{†,‡} MD 
Investigation performed at Institute of Sports Medicine, Department of Sports Medicine, Peking University Third Hospital, Beijing, China

Background: Individuals with anterior cruciate ligament (ACL) reconstruction (ACLR) are a population that has a higher risk for ACL injury compared with the general population. To reduce the reinjury rate and improve the rehabilitation outcome after ACLR, risk factors for ACL injury have to be addressed.

Purpose: To compare the knee biomechanics during side-cutting and isokinetic strength of the thigh muscle of the reconstructed leg with those of the contralateral leg and healthy controls and investigate the knee movement asymmetries in individuals with ACLR.

Study Design: Controlled laboratory study.

Methods: A total of 16 participants with ACLR (ACLR group; 11.8 ± 1.1 months after reconstruction) and 16 healthy controls (control group) were recruited. Landmark coordinates and ground-reaction forces during side-cutting and isokinetic strength of hamstring and quadriceps were collected. Two-way analysis of variance with the mixed design was performed to compare each dependent variable between groups and across legs.

Results: The reconstructed leg had a significantly smaller knee flexion angle ($P = .004$) and less quadriceps strength ($P = .003$) than the contralateral leg. The knee extension moment and knee external rotation angle were decreased compared with both the contralateral leg ($P = .001$, $P = .003$, respectively) and the healthy control leg ($P = .001$, $P = .001$, respectively). The ACLR group showed greater knee abduction angles ($P = .004$) and smaller knee external rotation moments ($P = .006$) than the control group. The ACLR group also demonstrated greater asymmetries of knee flexion angle ($P = .015$), knee external rotation angle ($P = .001$), knee extension moment ($P = .013$), knee abduction moment ($P = .001$), and quadriceps strength ($P = .046$) than the control group.

Conclusion: Knee biomechanics in the leg with ACLR were altered mainly in the sagittal plane during side-cutting compared with the contralateral leg. The altered movement patterns between the ACLR and control groups were primarily observed in the frontal and transverse planes. The ACLR group also demonstrated greater asymmetries of sagittal knee movement and concentric quadriceps strength than the control group.

Clinical Relevance: Individuals with ACLR showed different alterations in the reconstructed and contralateral leg compared with healthy controls. These results suggest that rehabilitation programs after ACLR should further focus on restoring the knee flexion angle and quadriceps strength. Injury prevention programs need to be further targeted in the altered movement patterns observed between the ACLR and the healthy groups.

Keywords: ACL reconstruction; cutting; asymmetry; biomechanics; strength

Anterior cruciate ligament (ACL) rupture is a common sports injury resulting in knee instability and early onset of knee osteoarthritis. ACL reconstruction (ACLR) surgery is a standard procedure to restore knee functions after an

ACL injury.⁷ Although ACLR surgeries generally restore knee stability, studies have shown that up to 12% of patients with ACLR have ACL reinjury.^{22,38} The secondary ACL injury rate in young athletes who return to sports after ACLR equates to 30 to 40 times that of uninjured athletes.⁴⁵

The high reinjury rate after ACLR is likely multifactorial, and various factors might affect a good outcome after ACLR, such as surgical technique, graft choice,

postoperative rehabilitation, and patient education. Unfortunately, graft failure and contralateral ACL rupture can still occur even after successful ACLR. A prospective study reported that the rate of ipsilateral retears and contralateral tears after ACLR was equal.⁴⁶ In another study, the overall rates of graft rupture and contralateral ACL injury were similar between patients who returned before and after 12 months postoperatively, and the subsequent ACL injury rate was 32% even for the patients who delayed their return.⁴⁴ A possible explanation of the increased risk in patients after ACLR is that these populations were actually at high risk for ACL injury before the surgery since a previous study reported that the relative risk of ACL injury may not change from preoperative to postoperative status.²⁹ To reduce the reinjury rate and improve the rehabilitation outcome after ACLR, risk factors for ACL injury have to be addressed.

Although some studies have explored the movement pattern and identified the altered knee kinematics and kinetics in individuals with ACLR, most of these studies investigated the biomechanics in lower-demand daily activities, such as walking and jogging.^{11,39} Side-cutting is a high-demand activity indicated as a crucial mechanism of ACL injury during sports.^{19,24} Thus, understanding the biomechanics during cutting movement is very important for exploring the ACL injury mechanism. Currently, only a few studies have investigated the biomechanics during side-cutting movement. Pollard et al³³ and Stearns and Pollard⁴² found that female soccer players with ACLR demonstrated altered knee biomechanics in the frontal plane during side-cutting compared with healthy controls. Recently, King et al¹⁸ also reported that patients with ACLR demonstrated more significant asymmetry of biomechanical variables during change-of-direction tasks compared with healthy controls. These studies mainly analyzed the reconstructed leg compared with healthy controls and did not pay much attention to the contralateral leg. As the rates of ipsilateral reinjury and contralateral injury after ACLR were similar, the common differences between the reconstructed leg and the legs of healthy individuals, as well as between the contralateral leg and the legs of healthy individuals, may be of more interest in identifying risk factors for ACL injury. In addition, previous studies have shown that the reconstructed leg demonstrates lower strength than the contralateral leg, emphasizing the importance of quadriceps and hamstring strength for knee movement control and functional

performance.^{2,3,27,37} Quadriceps symmetry has also been used as a prerequisite for a successful return to sports after ACLR.⁸ These studies used the contralateral leg as a reference and did not analyze whether the contralateral muscle strength was normal. Future studies should pay more attention to quadriceps and hamstring strength analysis, especially the differences between the contralateral leg in individuals with ACLR and the legs of healthy individuals.

Therefore, this study aims to evaluate and compare the knee kinematics and kinetics during side-cutting and isokinetic thigh muscle strength of the reconstructed leg against the contralateral leg and the legs of healthy controls, as well as investigate the knee movement asymmetries in individuals with ACLR. We hypothesized that there would be significant interleg differences in knee biomechanics during side-cutting of the individuals with ACLR. We also hypothesized that the contralateral leg would also have altered knee biomechanics and there would be significant differences between individuals with ACLR and healthy controls. We further hypothesized that the reconstructed leg would have significant strength deficits in the quadriceps and hamstring compared with the contralateral and healthy control legs.

METHODS

Participants

Sixteen male participants with unilateral ACL reconstruction (ACLR group) and 16 age-matched male healthy controls (control group) participated in this study (Table 1). Inclusion criteria for the ACLR group were as follows: isolated ACL injury without other ligament injuries, surgery using a hamstring tendon autograft, 1 year follow-up after ACLR, and no other current orthopaedic injuries or disorders that might affect lower extremity movements (also applied to control group). All participants with ACLR went through similar postoperative home-based rehabilitation programs.⁴⁰ All participants had a normal knee range of motion and were not limited when participating in the study. This study was approved by the institutional review board of Peking University Third Hospital. All participants read and signed a written informed consent form before data collection. The estimated effect size for knee abduction moments during side-cutting between individuals with ACLR and healthy individuals was 1.17 based on

[†]Address correspondence to Hui Liu, PhD, Biomechanics Laboratory, College of Human Movement Science, Beijing Sport University, 48 Xinxin Road, Haidian District, Beijing, 100084, P.R. China (email: liuhuibupe@163.com); Yingfang Ao, MD, Department of Sports Medicine, Peking University Third Hospital, Institute of Sports Medicine of Peking University, Beijing Key Laboratory of Sports Injuries, 49 North Garden Road, Haidian District, Beijing, 100191, P. R. China (email: aoyingfang@163.com).

*Biomechanics Laboratory, College of Human Movement Science, Beijing Sport University, Beijing, China.

[†]Department of Sports Medicine, Peking University Third Hospital, Institute of Sports Medicine of Peking University, Beijing Key Laboratory of Sports Injuries, Beijing, China.

H.S. and S.R. are co-first authors.

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TABLE 1
Participant Characteristics^a

Characteristic	ACLR Group, Mean $\pm$ SD	Control Group, Mean $\pm$ SD	P Value
Age, y	30.4 $\pm$ 4.7	29.4 $\pm$ 5.7	.602
Body mass, kg	82.1 $\pm$ 8.5	74.5 $\pm$ 11.9	.054
Body height, cm	177.1 $\pm$ 5.2	174.1 $\pm$ 8.0	.249
Time since surgery, mo	11.8 $\pm$ 1.1	NA	NA
IKDC 2000	82.3 $\pm$ 7.4	96.9 $\pm$ 4.7	<.001
Lysholm score	84.3 $\pm$ 11.9	99.5 $\pm$ 1.4	.001
Tegner score	5.1 $\pm$ 1.3	6.8 $\pm$ 1.5	.012

^aACLR, anterior cruciate ligament reconstruction; IKDC 2000, International Knee Documentation Committee Subjective Knee Form; NA, not applicable.

previous work.⁴² Therefore, we estimated that 13 individuals would be needed to achieve a power no less than 0.8 and a type I error no larger than .05 for group comparison (Version 3.1; G*Power). We recruited 16 patients and 16 healthy individuals. However, since 2 healthy individuals did not have time to undergo the isokinetic strength testing, only data from the remaining 14 individuals were used for the isokinetic strength analysis, while the kinematic and kinetic analysis during side-cutting used the data of all 16 individuals.

Protocol

Participants first completed the 2000 International Knee Documentation Committee Subjective Knee Form,¹⁶ Lysholm scale,⁴³ and Tegner activity scale.¹⁴ Then the participants were asked to complete the side-cutting test and isokinetic strength test. Before the side-cutting test, each participant was asked to wear a pair of black spandex shorts. Passive reflective markers were placed bilaterally at the anterior superior iliac spine, posterior superior iliac spine, lateral thigh, lateral femoral condyle, medial femoral condyle, anterior superior shank, anterior inferior shank, lateral malleolus, medial malleolus, heel, and first and fifth metatarsophalangeal joints. In a side-cutting test, the participant was instructed to complete the movement as fast as he was comfortable doing.⁵ In the ACLR group, the contralateral leg was tested first and then the reconstructed leg. In the control group, the dominant leg was tested first and then the nondominant leg. Dominant leg was defined as the leg regularly used to kick a ball. Each participant had 3 successful trials for each leg in the side-cutting test. A successful trial was when the participant completed the task as required and investigators collected all kinematic and kinetic data. Thigh muscle strength was evaluated for the participant after the side-cutting test.

Data Collection

In a side-cutting test, 3-dimensional trajectories of the reflective markers were collected using an 8-camera motion capture system (Vicon) at a sample rate of 100 Hz. Ground-

reaction force (GRF) signals were collected using 2 embedded force plates (AMTI) at a sample rate of 1000 Hz.

In an isokinetic test, quadriceps and hamstring muscle strength were quantified using an isokinetic dynamometer (CON-TREX MJ) at a sample rate of 256 Hz. The participant was seated on the strength testing system with the trunk and testing thigh fixed on the system using belts. The knee range of motion was fixed from 90° of flexion to 20° of flexion with an angular velocity of 60 deg/s, which is the angular velocity commonly used in individuals who underwent ACLR,^{6,10} as it has been reported to reliably assess knee extension and flexion strength.⁴¹ Each participant was asked to perform 3 preliminary submaximal repetitions to familiarize themselves with the testing apparatus before each test. The strength assessment protocol was as follows: 5 maximal tests in concentric mode for the quadriceps and hamstring at 60 deg/s (3 trials) and 5 maximal tests in eccentric mode for the quadriceps and hamstring at 60 deg/s (3 trials). All trials of tests were separated by 3 minutes of rest.

Data Reduction

The raw trajectories of the reflective markers were filtered via a Butterworth low-pass digital filter at an estimated optional cutoff frequency of 10 Hz.⁴⁷ Kinematic and kinetic variables were calculated using Visual 3D software (C-Motion). Knee joint angles were calculated as Cardan angles of a distal segment reference frame relative to the proximal segment reference frame in order of (1) flexion-extension, (2) abduction-adduction, and (3) internal-external rotation. Joint moments were calculated using an inverse dynamic approach and transferred into the distal segment reference frame. GRFs were normalized to the body weight (BW). Joint moments and muscle strength were normalized to the production of BW (measured in newtons) and standing height (BH; measured in meters) and calculated as BW * BH.

The peak posterior GRF (pGRF), the peak vertical GRF, and the peak torque in the isokinetic strength assessment were identified for each participant in each trial. Since peak pGRF and peak proximal tibial anterior shear force during the landing were significantly correlated and occurred essentially at the same time,⁴⁸ the knee

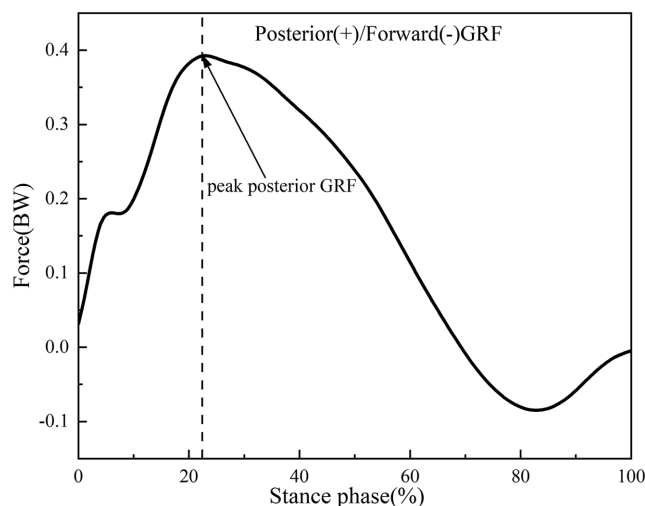


Figure 1. The method for determining peak posterior ground-reaction force (GRF) using force data in the posterior-forward direction measured via the force plate. BW, body weight.

kinematics and kinetics at peak pGRF during side-cutting were also identified in each trial for each participant. The method for determining the pGRF in this study is illustrated in Figure 1.

In addition to the above discrete measures, this study also compared the symmetry of bilateral knee biomechanics and muscle strength between the ACLR and control groups. The knee kinematics and kinetics asymmetry between legs were evaluated using interleg difference (ILD). The thigh muscle strength asymmetry between legs was evaluated using the limb symmetry index (LSI). The ILD³⁹ between legs in each group was calculated for joint angles and moments using

$$ILD = Y_{leg1} - Y_{leg2},$$

where Y_{leg1} and Y_{leg2} were magnitudes of a given joint angle or moment of the contralateral leg and reconstructed leg in the ACLR group or of the dominant leg and nondominant leg in the control group.

The LSI was calculated for GRF and thigh muscle strength using

$$LSI = \frac{Y_{leg1}}{Y_{leg2}} * 100\%,$$

where Y_{leg1} and Y_{leg2} were the magnitudes of the GRF and isokinetic strength of the reconstructed leg and contralateral leg in the ACLR group or of the nondominant leg and dominant leg in the control group.

Statistical Analysis

All variables were tested for normality using the Kolmogorov-Smirnov test and found to be normally distributed. Two-way analysis of variance (ANOVA) with the mixed design was performed to compare each dependent

variable between groups (ACLR and control groups) and across legs (reconstructed and contralateral legs for the ACLR group, dominant and nondominant legs for the control group). ACL injury status was the between-participant factor, and leg was the within-participant factor. If a significant interaction effect of group and leg was detected, a total of 4 *t* tests were performed, including 2 paired *t* tests within each group (reconstructed leg vs contralateral leg within the ACLR group; dominant leg vs nondominant leg within the control group) and 2 independent *t* tests between groups (reconstructed leg in the ACLR group vs nondominant leg in the control group; contralateral leg in the ACLR group vs dominant leg in the control group). The Bonferroni procedure was used to adjust the type I error rate to .0125 as statistical significance to ensure that the overall type I error rate was no greater than .05. Independent *t* tests were performed to compare the ILD of knee kinematics and kinetics and the LSI of GRF and isokinetic strength between the ACLR group and the control group. All data analyses were performed using the SPSS computer program package (Version 16.0; SPSS, Inc). We also defined minimal clinically important differences for quadriceps strength (0.186 BW * BH), knee flexion angle (3 deg), knee rotation angle (3 deg), and knee moment (0.004 BW * BH) to determine whether the results of our statistical comparisons were clinically meaningful.^{32,35}

RESULTS

Knee Biomechanics

Two-way ANOVA revealed significant interaction effects of group and leg on the knee flexion angle and knee rotation angle at peak pGRF ($P = .015$, $P < .001$, respectively) (Table 2). The knee flexion angle and knee rotation angle at peak pGRF of the reconstructed leg were significantly smaller than those of the contralateral leg in the ACLR group ($P = .004$, $P = .003$, respectively) (Figure 2A), whereas no significant differences were found between legs in the control group ($P = .765$, $P = .032$, respectively). The knee flexion angles at peak pGRF were similar between the reconstructed and the nondominant legs ($P = .020$), and between the contralateral and the dominant legs ($P = .695$). The reconstructed leg had significantly greater knee rotation angle at peak pGRF than the nondominant leg in the control group ($P = .001$) (Figure 2C), but no significant difference was observed between the contralateral leg in the ACLR group and the dominant leg in the control group ($P = .228$). The ILDs of the knee flexion angle and knee external rotation angle at peak pGRF in the ACLR group were significantly greater than those in the control group ($P = .015$, $P < .001$, respectively) (Table 3). The differences in knee flexion angle and knee rotation angle at peak pGRF between the reconstructed leg and the contralateral leg and between the reconstructed leg and the control leg were clinically meaningful.

No significant interaction effect of group and leg on knee abduction angle at peak pGRF was detected ($P = .684$) (Table 2, Figure 2B). Knee abduction angle at peak pGRF in the ACLR group was significantly greater

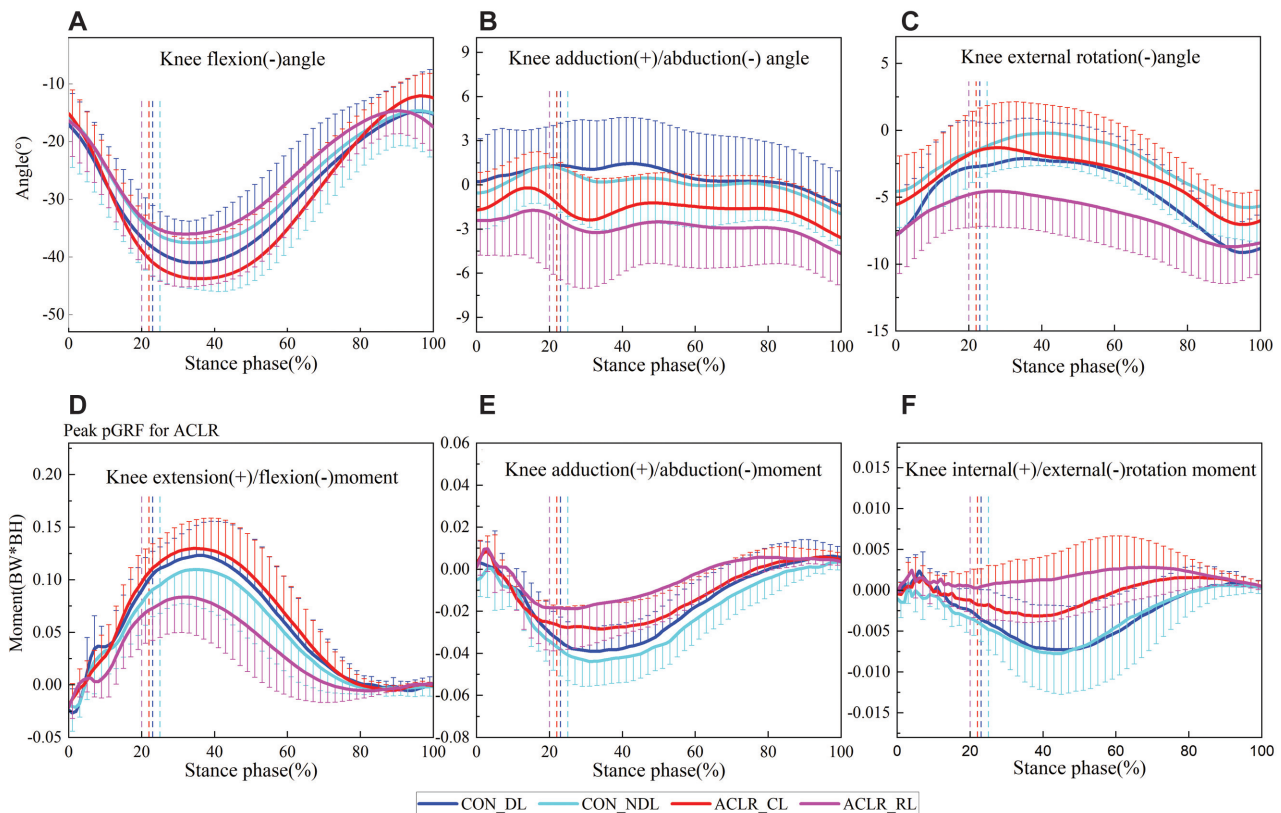


Figure 2. Knee angles in the sagittal (A), coronal (B), and horizontal (C) planes and moments in the sagittal (D), coronal (E), and horizontal (F) planes during side-cutting. ACLR_CL, the contralateral leg in the ACLR group; ACLR_RL, the reconstructed leg in the ACLR group; BW * BH, body weight multiplied by body height; CON_DL, the dominant leg in the control group; CON_NDL, the nondominant leg in the control group. The vertical lines represent the instant of peak posterior ground-reaction force (pGRF) for each curve (blue, the dominant leg in the control group; cyan, the nondominant leg in the control group; red, the contralateral leg in the ACLR group; magenta, the reconstructed leg in the ACLR group). ACLR, anterior cruciate ligament reconstruction.

compared with that of the healthy control group ($P = .004$). No significant leg effect was observed in knee abduction angle at peak pGRF ($P = .239$). The ILD of the knee abduction angle at peak pGRF between the ACLR group and the control group was similar ($P = .684$) (Table 3).

Significant interaction effects of group and leg on knee extension moment and knee abduction moment at peak pGRF were detected ($P = .013$, $P = .001$, respectively) (Table 2). In the ACLR group, the knee extension moment at peak pGRF of the reconstructed leg was significantly reduced compared with that of the contralateral leg ($P = .001$) (Figure 2D). However, in the control group, the knee extension moment was similar between legs ($P = .802$). No significant difference in knee abduction moment between legs was observed in both ACLR group ($P = .014$) and control group ($P = .042$) (Figure 2E). The knee extension moment and knee abduction moment at peak pGRF of the reconstructed leg in the ACLR group were significantly reduced compared with that of the nondominant leg in the control group ($P = .001$, $P = .001$, respectively). However, no significant differences were observed between the contralateral leg in the ACLR group and the dominant leg in the control group ($P = .610$, $P = .043$, respectively). The ILDs of the knee extension

moment and knee abduction moment at peak pGRF in the ACLR group were significantly greater than those in the control group ($P = .013$, $P = .001$, respectively) (Table 3).

No significant interaction effects of group and leg on knee rotation moment at peak pGRF were detected ($P = .085$) (Table 2). Knee external rotation moment at peak pGRF in the ACLR group was significantly smaller than that in the control group ($P = .006$) (Figure 2F). Significant leg effect was observed in knee external rotation moment at peak pGRF ($P = .049$). The ILD of knee external rotation moment in the ACLR group was similar to that in the control group ($P = .085$) (Table 3).

A significant interaction effect of group and leg on peak vertical GRF was detected ($P = .013$) (Table 2). The peak vertical GRF of the reconstructed leg was significantly smaller than that of the contralateral leg in the ACLR group ($P = .001$), while no significant difference was observed between legs in the control group ($P = .546$). The peak vertical GRF of the reconstructed and contralateral legs in the ACLR group was similar to that of the nondominant and dominant legs in the control group ($P = .065$, $P = .974$, respectively). No significant interaction effect of group and leg on peak pGRF were detected ($P = .885$)

TABLE 2
Knee Biomechanics at Peak Posterior Ground-Reaction Force^a

Characteristic	ACLR Group, Mean $\pm$ SD		Control Group, Mean $\pm$ SD		P Value		
	Reconstructed Leg	Contralateral Leg	Nondominant Leg	Dominant Leg	Interaction Effect	Between Leg	Between Group
Knee flexion angle, deg	29.8 $\pm$ 9.9	35.5 $\pm$ 9.5	37.3 $\pm$ 7.1	36.8 $\pm$ 7.7	.015 ^b	.004 ^{b,c}	.020 ^d
Knee adduction (+)/abduction (–) angle, deg	–2.3 $\pm$ 3.8	–1.6 $\pm$ 3.5	1.1 $\pm$ 2.7	1.4 $\pm$ 2.6	.684	.765 ^e	.695 ^f
Knee external rotation angle, deg	4.7 $\pm$ 2.6	1.4 $\pm$ 3.7	1.3 $\pm$ 2.6	3.0 $\pm$ 3.8	<.001 ^b	.003 ^{b,c}	.001 ^{b,d}
Knee extension moment, BW * BH	0.057 $\pm$ 0.033	0.092 $\pm$ 0.037	0.096 $\pm$ 0.027	0.098 $\pm$ 0.032	.013 ^b	.001 ^{b,c}	.001 ^{b,d}
Knee abduction moment, BW * BH	0.012 $\pm$ 0.025	0.023 $\pm$ 0.014	0.041 $\pm$ 0.014	0.033 $\pm$ 0.009	.001 ^b	.014 ^c	.001 ^{b,d}
Knee external rotation moment, BW * BH	0.001 $\pm$ 0.004	0.002 $\pm$ 0.004	0.004 $\pm$ 0.002	0.004 $\pm$ 0.003	.085	.049 ^b	.006 ^b
Peak vertical GRF, BW	1.967 $\pm$ 0.156	2.138 $\pm$ 0.202	2.117 $\pm$ 0.271	2.141 $\pm$ 0.272	.013 ^b	.001 ^{b,c}	.065 ^d
Peak posterior GRF, BW	0.380 $\pm$ 0.071	0.427 $\pm$ 0.069	0.382 $\pm$ 0.071	0.425 $\pm$ 0.100	.885	.546 ^e	.974 ^f

^aACLR, anterior cruciate ligament reconstruction; BW, body weight; BW * BH, body weight multiplied by body height; GRF, ground-reaction force.

^bIndicates significant interaction and main effects.

^cIndicates comparison between reconstructed and contralateral legs in the ACLR group.

^dIndicates comparison between reconstructed leg in the ACLR group and nondominant leg in the control group.

^eIndicates comparison between nondominant and dominant legs in the control group.

^fIndicates comparison between contralateral leg in the ACLR group and dominant leg in the control group.

TABLE 3
Asymmetry Index of Knee Biomechanics and Isokinetic Strength^a

Variable	ACLR Group, Mean $\pm$ SD	Control Group, Mean $\pm$ SD	P Value
ILD			
Knee flexion angle at pGRF, deg	5.7 $\pm$ 6.5	0.5 $\pm$ 6.9	.015 ^b
Knee adduction angle at pGRF, deg	0.7 $\pm$ 2.9	0.4 $\pm$ 1.9	.684
Knee external rotation angle at pGRF, deg	3.3 $\pm$ 3.6	1.7 $\pm$ 2.9	<.001 ^b
Knee extension moment at pGRF, BW * BH	0.034 $\pm$ 0.032	0.002 $\pm$ 0.035	.013 ^b
Knee abduction moment at pGRF, BW * BH	0.011 $\pm$ 0.016	0.008 $\pm$ 0.015	.001 ^b
Knee external rotation moment at pGRF, BW * BH	0.003 $\pm$ 0.004	0.001 $\pm$ 0.003	.085
LSI, %			
Peak vertical GRF, BW	92.4 $\pm$ 6.8	99.1 $\pm$ 7.1	.012 ^b
Peak posterior GRF, BW	89.9 $\pm$ 15.4	91.9 $\pm$ 15.8	.719
Qc	76.0 $\pm$ 24.3	96.3 $\pm$ 20.8	.046 ^b
Hc	86.1 $\pm$ 13.0	93.2 $\pm$ 15.0	.231
Qe	85.2 $\pm$ 20.2	96.1 $\pm$ 17.8	.191
He	85.1 $\pm$ 15.8	91.6 $\pm$ 19.1	.371

^aACLR, anterior cruciate ligament reconstruction; BW, body weight; BW * BH, body weight multiplied by body height; GRF, ground-reaction force; ILD, interleg difference; LSI, leg symmetry index; pGRF, peak posterior ground-reaction force; Hc, concentric hamstring strength; He, eccentric hamstring strength; Qc, concentric quadriceps strength; Qe, eccentric quadriceps strength.

^bIndicates significant effects between groups.

(Table 2). A significant leg effect in peak pGRF was found ($P = .002$). No significant difference in peak pGRF was observed between groups ($P = .996$). The LSI of the peak pGRF was similar between the ACLR group and the control group ($P = .719$). In contrast, the LSI of the peak vertical GRF in the ACLR group was significantly smaller than that in the control group ($P = .012$).

Isokinetic Strength

A significant interaction effect of group and leg on peak concentric quadriceps strength (Qc) was detected ($P =$

.045) (Table 4, Figure 3A). In the ACLR group, the peak Qc of the reconstructed leg was significantly less than that of the contralateral leg ($P = .003$). However, in the control group, the peak Qc was similar between legs ($P = .688$). The peak Qc was similar between the reconstructed and the nondominant legs ($P = .104$), and between the contralateral and the dominant legs ($P = .589$) to that of both legs in the control group ($P = .104$, $P = .589$). The individuals with ACLR demonstrated greater LSI of Qc than the healthy controls ($P = .046$) (Table 3).

No significant interaction effects of group and leg on peak concentric hamstring strength (Hc), peak eccentric

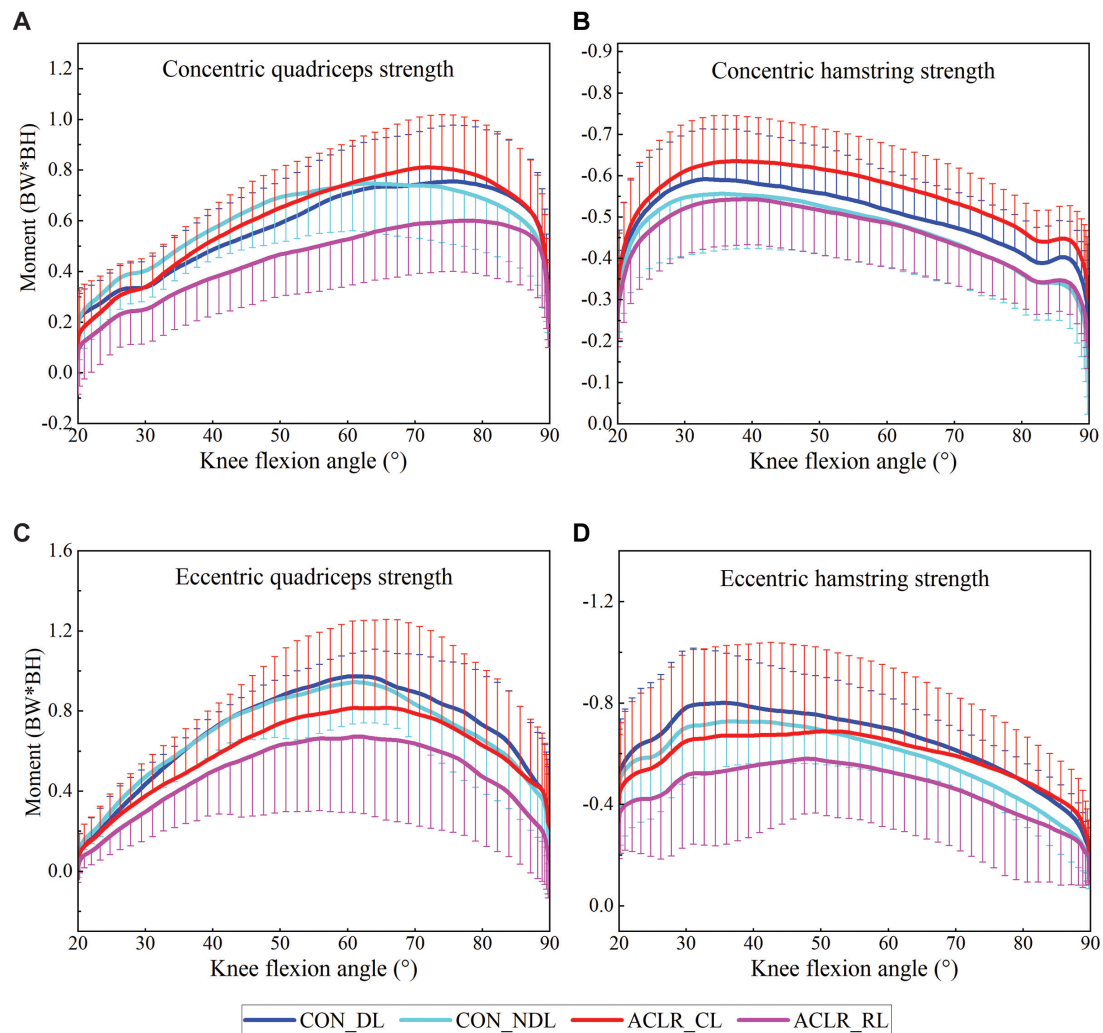


Figure 3. Isokinetic strength of the quadriceps in concentric (A) and eccentric (C) contraction modes and the hamstring in concentric (B) and eccentric (D) contraction modes. ACLR, anterior cruciate ligament reconstruction; ACLR_CL, the contralateral leg in the ACLR group; ACLR_RL, the reconstructed leg in the ACLR group; BW * BH, body weight multiplied by body height; CON_DL, the dominant leg in the control group; CON_NDL, the nondominant leg in the control group.

TABLE 4
Peak Isokinetic Quadriceps and Hamstring Strength^a

Characteristic	ACLR Group, Mean ± SD		Control Group, Mean ± SD		P Value		
	Reconstructed Leg	Contralateral Leg	Nondominant Leg	Dominant Leg	Interaction Effect	Between Leg	Between Group
Qc, BW * BH	0.64 ± 0.22	0.87 ± 0.21	0.80 ± 0.20	0.82 ± 0.21	.045 ^b	.003 ^{b,c}	.104 ^d
Hc, BW * BH	0.57 ± 0.11	0.66 ± 0.11	0.58 ± 0.14	0.63 ± 0.11	.209	.688 ^e	.589 ^f
Qe, BW * BH	0.87 ± 0.24	1.04 ± 0.24	0.98 ± 0.20	1.06 ± 0.09	.283	.001 ^b	.837
He, BW * BH	0.68 ± 0.15	0.83 ± 0.21	0.76 ± 0.17	0.88 ± 0.20	.603	.003 ^b	.421
						<.001 ^b	.358

^aACLR, anterior cruciate ligament reconstruction; BW * BH, body weight multiplied by body height; Hc, concentric hamstring strength; He, eccentric hamstring strength; Qc, concentric quadriceps strength; Qe, eccentric quadriceps strength.

^bIndicates significant interaction and main effects.

^cIndicates comparison between the reconstructed and contralateral legs in the ACLR group.

^dIndicates comparison between the reconstructed leg in the ACLR group and nondominant leg in the control group.

^eIndicates comparison between the nondominant and dominant legs in the control group.

^fIndicates comparison between the contralateral leg in the ACLR group and dominant leg in the control group.

quadriceps strength (Qe), and peak eccentric hamstring strength (He) were detected ($P = .209$, $P = .283$, $P = .603$, respectively) (Table 4, Figure 3B, C, and D). Peak Hc, Qe, and He in the ACLR group were similar to that in the control group ($P = .837$, $P = .421$, $P = .358$). Significant leg effects were observed in peak Hc, Qe, and He ($P = .001$, $P = .003$, $P < .001$), but the differences between legs in each group were not clinically meaningful. The LSIs of Hc, Qe, and He were similar between the individuals with ACLR and healthy controls ($P = .231$, $P = .191$, $P = .371$, respectively) (Table 3).

DISCUSSION

The results of this study supported our hypothesis that there would be significant ILDs in knee biomechanics during side-cutting of the individuals with ACLR. This study found that the reconstructed leg had significantly decreased knee flexion angle compared with the contralateral leg, which is consistent with the literature. Gokeler et al⁹ found that the knee flexion angle on the reconstructed leg was significantly smaller during landing. Dai et al⁵ also found that the knee flexion angle of the reconstructed leg was smaller than that of the contralateral leg during side-cutting. A smaller knee flexion angle would increase the tilt angle of the ACL. Thus, the anterior tibial shear force generated by the quadriceps would be greater.^{20,23,48} Although studies have repeatedly recommended that injury prevention programs focus on landing with greater knee flexion angle,²¹ individuals with ACLR continue to adopt these potentially deleterious movement patterns. In addition, the current study also observed that individuals with ACLR demonstrated greater ILD of the knee flexion angles than control participants, which suggests that individuals with ACLR have asymmetrical knee flexion movement between legs. Previous studies have repeatedly reported asymmetric knee flexion movements after ACLR, which is considered a risk factor for primary ACL injury and injury after ACLR.^{15,18} The present study found the knee flexion angle of the reconstructed leg was smaller during side-cutting movement than that of the contralateral leg in the ACLR group, but no significant difference was observed between the reconstructed leg and the healthy control leg or between the contralateral leg and the healthy control leg, suggesting that the smaller knee flexion angle of the reconstructed leg may be due to the surgery. The current study indicated that the reduced knee flexion angle of the reconstructed leg during side-cutting movement is the essential manifestation of abnormal characteristics and it is also the root cause of the asymmetry of the knee flexion angle. The current results, combined with the results of previous studies, indicated that the existing postoperative rehabilitation program may not be adequate for restoring the knee flexion movement pattern and that a specific rehabilitation program needs to be designed subsequently.

The results of this study supported our hypotheses that the contralateral leg would also alter knee biomechanics

and there would be significant differences between individuals with ACLR and healthy controls. The present study found that the differences between the individuals with ACLR and healthy controls were observed mainly in the frontal plane. The present study showed that individuals with ACLR demonstrated a greater knee abduction angle at peak pGRF than healthy controls, which is consistent with the literature. A previous study found that female soccer players with ACLR showed increased knee abduction angle during side-cutting compared with healthy female soccer players.⁴² A prospective cohort study¹³ reported that individuals with ACL injury demonstrated significant increases in dynamic valgus before sustaining their injuries compared with uninjured controls, concluding that increased dynamic valgus is an important risk factor for ACL injury. Some other researchers used the lower extremity cadaveric specimens simulating landing after a jump to investigate the relationship between the knee abduction movement and ACL injury¹⁷ and have found that the degree of knee valgus was significantly associated with the ACL strain, suggesting that knee valgus is a significant risk factor for noncontact injury. Clark et al⁴ also reported that 60% of the uninjured legs in athletes with ACLR also exhibited greater knee abduction angles. In this study, they concluded that it might not be reasonable to use the uninjured leg as a criterion for rehabilitation on the reconstructed leg. Combining the results of previous studies, the current study further demonstrated that not only the reconstructed leg but also the contralateral leg had significantly increased knee abduction angle compared with the healthy control legs, which could reveal the deficits in the contralateral leg in the individuals with ACLR. The intergroup difference in the increased knee abduction angle is likely to be an essential risk factor for reinjury on both the reconstructed and contralateral legs of the individuals with ACLR.

This study also demonstrated that the knee external rotation angle at peak pGRF of the reconstructed leg was greater than that of both the contralateral leg in the ACLR group and the nondominant leg in the control group. Significant leg-to-leg differences in knee external rotation angle were observed in individuals with ACLR but not in healthy controls. Scanlan et al³⁶ conducted a walking study in patients with ACLR and found that the knee external rotation angle of the reconstructed leg was significantly greater than that of the uninjured leg. The ACL limits anterior tibial displacement and plays a vital role in limiting excessive knee rotation. The greater external rotation angle of the knee on the reconstructed leg may suggest that the rotational stability of the lower extremity is not sufficiently controlled after ACLR. As the larger rotational position of the knee would increase the ACL loading, the restoration of knee rotational stability remains a focus of clinical rehabilitation and ACL injury prevention.

The results of this study showed that the reconstructed leg had a smaller knee extension moment at peak pGRF compared with both the contralateral leg and the healthy control leg, while no significant difference between legs was found in healthy controls. The differences in knee

extension moment between the reconstructed leg and the contralateral leg and between the reconstructed leg and the control leg were clinically meaningful since the differences were greater compared with the minimal clinical difference ($0.004 \text{ BW} \cdot \text{BH}$).³⁵ The present study also found that the ILD of the ACLR group was significantly greater than that of healthy controls. A recent review also reported that the knee extension moment of the reconstructed leg during running was significantly reduced whether it was compared with the contralateral leg or the healthy control leg.²⁶ Joint moments represent muscle demand around the joint. Since the knee flexion angle of the reconstructed leg was also smaller, this smaller extension moment may be due to less need for eccentric contraction of the knee extensor muscles in the more extended position. In addition, consistent with previous studies,^{12,25} the current study also found that the quadriceps strength of the reconstructed leg was weaker than that of the contralateral leg, so the lower knee extension moment may also be caused by insufficient knee extension muscle strength. However, since quadriceps strength deficits after ACLR are a chronic feature, it is also possible that the abnormal movement pattern results in lower voluntary activation in the reconstructed leg, resulting in long-term disuse atrophy of quadriceps strength.²⁵ It should also be emphasized here that because the present study was a cross-sectional study, it was not possible to determine whether the abnormal movement pattern alteration resulted in quadriceps strength deficits or the weaker quadriceps strength led to the lower knee extension moment. Future studies should further investigate the causal relationship between specific alterations in movement patterns and reduced muscle strength.

The present study also found that a significant difference in the peak vertical GRF was present between legs in individuals with ACLR; that is, the reconstructed leg demonstrated a significantly lower peak vertical GRF compared with the contralateral leg. The decreased peak vertical GRF on the reconstructed leg may result from the compensatory pattern of unloading the reconstructed leg during a dynamic task. It may also be due to the psychological lack of confidence in the reconstructed leg. Whether the observed difference was based on an adapted movement pattern and/or psychological confidence remains unclear and needs further research. Paterno et al³⁰ identified similar unloading of the reconstructed leg compared with the contralateral leg within athletes 2 years after ACLR. In addition, the current study also found that the LSI of the vertical GRF in the ACLR group was significantly smaller than that in the control group. The limb asymmetries in the vertical GRF during a jump landing task were also identified in a previous study.²⁸ The peak vertical GRF asymmetry was considered the most prominent risk factor associated with a noncontact lower extremity injury.³⁴ The current study, combined with the literature, indicates that landing force asymmetry is a potential risk factor for ACL injury. The intervention programs could focus on monitoring the state of loading patterns.

Concerning thigh muscle strength, the results of this study partially support the hypothesis that the

reconstructed leg would have significant strength deficits in quadriceps and hamstring compared with both the contralateral and the healthy control legs. This study demonstrated that the reconstructed leg had a smaller peak Qc compared with the contralateral leg, while no significant ILD was observed in the healthy control group. The differences in peak Qc between the reconstructed leg and the contralateral leg were clinically meaningful since the differences were greater compared with the minimal clinical difference ($0.186 \text{ BW} \cdot \text{BH}$).³² This study further showed that the individuals with ACLR demonstrated increased LSI of Qc compared with healthy controls. In the present study, similar leg symmetry of Hc, Qc, and He was observed between the ACLR group and the control group. No significant differences in these variables were found between the ACLR group and the control group. These results suggested that only Qc showed significant deficiency and asymmetry in individuals with ACLR. A 2-year prospective study¹⁰ showed that quadriceps strength deficit before returning to sports increased the risk of knee reinjury. More symmetrical quadriceps strength before returning to sports significantly reduced the knee reinjury rate. Due to the large effect that quadriceps strength may have on knee function, isolated training to increase quadriceps strength has been highly recommended in rehabilitation¹ to limit alterations during explosive types of activities such as cutting and jumping and improve long-term outcomes after ACLR. Future research should focus on more specific and effective interventions to improve quadriceps strength.

The findings of this study have important clinical implications for post-ACLR rehabilitation programs and reinjury prevention. As altered movement control patterns during dynamic activities and deficits in postural stability have been implicated as a predictor of ACL reinjury,³¹ it is crucial to restore biomechanics during dynamic activities. Indicators that differ between the reconstructed leg and the contralateral leg, but not between the contralateral leg and the dominant leg in healthy controls, such as knee flexion angle at peak pGRF and quadriceps strength, may only reflect deficits in knee function due to reconstructive surgery and may not indicate that these differential variables are risk factors for ACL injury. The differences between the ACLR and control groups reflected bilateral deficits, which may better indicate the risk factors for injury in individuals with ACLR. These deficits indicate that residual impairments with dynamic activities still persist 1 year after surgery, potentially increasing risk of future injury on both the reconstructed and the contralateral legs. The increased knee abduction angle at pGRF and the decreased knee external rotation moment should be targeted not only for the reconstructed leg but also for the contralateral leg in future injury prevention programs. In clinics, although comparison with the contralateral leg could monitor the recovery after ACLR, compensation by the contralateral leg or bilateral deficits may influence the comparison between the reconstructed leg and the contralateral leg after ACLR. Researchers should be aware of this limitation, and it is recommended that studies exploring ACL injury risk factors based on individuals with

ACLR should focus more on intergroup comparisons with healthy individuals.

Further studies are needed to investigate whether targeted training in individuals after ACLR helps restore the altered biomechanics (reduced knee angles and moments) during high-demand tasks, including cutting and jumping. Future studies should also further confirm whether recovery of these differential biomechanical factors can actually reduce the injury rate. Although the results of this study demonstrated that individuals with ACLR altered knee biomechanics, questions remain as to the clinical importance of the differences in rotational motion, and follow-up studies are required to determine if these differential alterations here are clinically meaningful.

CONCLUSION

The results of this study warrant the following conclusions:

1. The reconstructed leg had a smaller knee flexion angle and quadriceps strength during side-cutting than the contralateral leg, and it showed decreased knee extension moment and increased knee external rotation angle compared with both the contralateral leg and the healthy control leg.
2. The asymmetries of the knee flexion and rotation angle, knee extension moment, and quadriceps strength in the participants with ACLR were significantly greater than those in healthy controls.
3. Compared with the healthy control group, the participants with ACLR demonstrated greater knee abduction angle and knee external rotation moment, as well as greater asymmetries of the knee extension moment and knee external rotation angle.

ORCID iD

Yingfang Ao  <https://orcid.org/0000-0002-8909-2022>

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